Name: ____________________ Reg #: ____________________ Section: ______________

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence **Course Code: AI2002**
Program: BS(Computer Science) **Semester: Spring 2026** **Section: BCS-6C**
Duration: 30 Minutes **Total Marks: 15**
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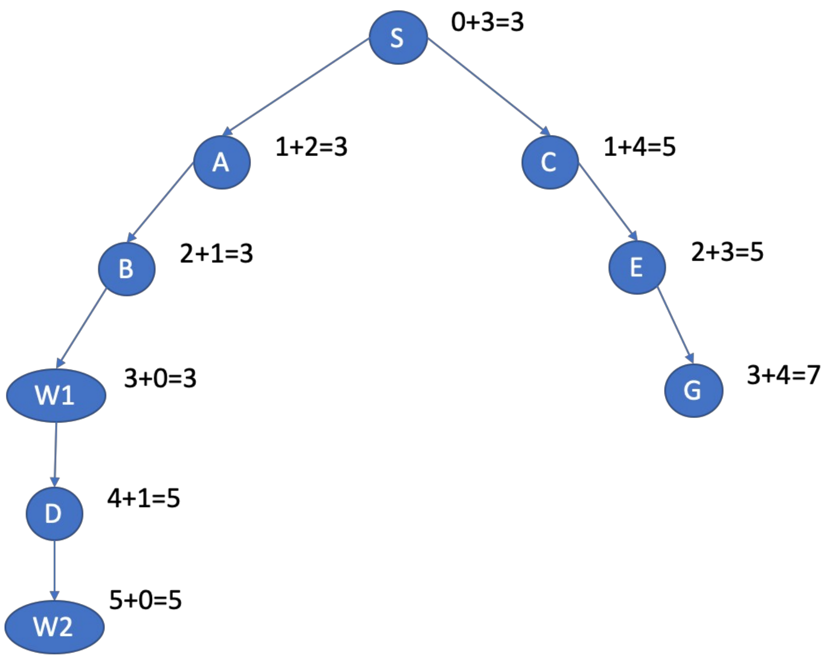
Exam: Quiz 2

Q1) A rover on Mars is exploring a rocky valley. It starts from its landing site and must reach one of two water deposits first, then continue to find the second. Some terrain is blocked by deep craters.

S	A	B	**W1**
C	X	X	D
E	X	F	**W2**
G	H	I	J

- S = Rover start coordinates (1,1)
- W1 and W2 = Water sources (Goals) with coordinates (1,4) and (3,4) respectively
- X = Craters (blocked)
- Move cost = 1
- Rover moves up, down, left, right
- The rover uses A* search with Manhattan Distance to the nearest goal as a heuristic.
  h(n) = |x1-x2|+|y1-y2|

(a) Draw the search tree Using A*. Calculate and write down the evaluation cost of each node. (4)



(b) Write the order in which nodes are expanded. (1)

S(3), A(3), B(3), W1(3), C(5), D(5), E(5)

(c) Must maintain both the frontier (open list) and the visited (closed list). (4)

Open / Frontier	Closed / Visited
S(3)	-
A(3),C(5)	S(3)
B(3), C(5)	A(3), S(3)
W1(3), C(5)	B(3), A(3), S(3)
C(5), D(5)	W1(3), B(3), A(3), S(3)
D(5), E(5)	C(5), W1(3), B(3), A(3), S(3)
E(5), W2(5)	D(5), C(5), W1(3), B(3), A(3), S(3)
W2(5), G(7)	E(5), D(5), C(5), W1(3), B(3), A(3), S(3)

(d) Find the final solution path in which both water sources are explored exactly once. (1)

S(3), A(3), B(3), W1(3), D(5), W2(5)

(e) Check whether the heuristic used are admissible. (4)

n	H(n)	Actual Cost	Admissible?
S	3	3	Yes
A	2	2	Yes
B	1	1	Yes
W1	0	0	Yes
C	4	4	Yes
D	1	1	Yes
E	3	4	Yes
F	1	1	Yes
W2	0	0	Yes
G	4	4	Yes
H	3	3	Yes
I	2	2	Yes
J	1	1	Yes

(f) When the solution is found, mention the states that remain in the frontier. (1)

W2(5), G(7)